Data Centers and Local Labor Markets

Dany Bahar[a,b] and Greg C. Wright[c]

[a]Center for Global Development, 2055 L Street NW, Fifth Floor, Washington, DC 20036, USA
[b]Brown University, Providence, RI 02912, USA
[c]University of California, Merced, 5200 North Lake Rd., Merced, CA 95343, USA

Dany Bahar: dbahar@cgdev.org
Greg C. Wright: gwright4@ucmerced.edu
Corresponding author: Greg C. Wright

April 17, 2026

Abstract

We construct a dataset of approximately 770 dated US data center facilities and estimate their local employment effects using synthetic control methods. Total private employment rises by 4–5 percent and information-sector employment by 22 percent. The information-sector effect is concentrated in counties hosting hyperscale (cloud and AI) operators: a triple-difference specification yields a hyperscale–colocation differential of 31 percentage points, and within a size-matched sample the same split produces a 23-percentage-point divergence in information-sector wages but no divergence in aggregate wages. Census QWI data confirm that both incumbent workers and new hires see earnings gains, indicating labor market tightening. These results imply that the local incidence of digital infrastructure investment depends sharply on facility type, with policy implications for the uniform tax treatment currently applied to hyperscale and colocation projects.

JEL: H25, J23, R11, R23
Keywords: data centers, local labor markets, hyperscale, colocation, agglomeration

1 Introduction

Place-based capital investment sometimes generates sizable local agglomeration and sometimes does not. The Tennessee Valley Authority raised manufacturing employment for half a century [Kline and Moretti, 2014], large plant openings lifted incumbent productivity in their host counties [Greenstone et al., 2010], and state corporate tax cuts passed through primarily to firm owners rather than workers [Suárez Serrato and Zidar, 2016]. Hydraulic fracturing produced local income gains tightly bounded by housing-price capitalization [Bartik et al., 2019], and state business incentives deliver widely varying cost-per-job depending on whom they target [Slattery and Zidar, 2020, Bartik, 2020]. The unresolved question is not whether a local multiplier exists, but *when*: what characteristics of a capital shock determine whether the host labor market captures meaningful spillovers.

This paper argues that the identity of the investing firm, and in particular whether it anchors a local supply chain of specialized inputs, is a first-order determinant of the multiplier, and that capital intensity *per se* is not. We make that case using data centers as a laboratory. Data centers are an attractive setting for two reasons. First, they sit at an extreme point on the capital-to-labor ratio spectrum, so whatever multiplier they generate cannot be attributed to sheer headcount at the facility itself. Second, the industry splits cleanly into two operator types that hold capital intensity roughly fixed but differ sharply in the local ecosystem they sustain: *hyperscale* operators (Amazon, Google, Microsoft, Meta) build campuses to run their own cloud and AI workloads and procure fiber construction, managed services, and network operations locally, while *colocation* operators (Equinix, Digital Realty) build landlord facilities leased to remote tenants whose inputs are not sourced in the host county. The contrast lets us hold the capital shock nearly fixed while varying the scope for local supplier linkages.

The empirical stakes are substantial. The United States now hosts more than 1,200 large data center facilities, annual investment in new capacity exceeds $100 billion, and a single hyperscale campus typically costs $500 million to $2 billion, draws 100–500 megawatts of continuous electrical load, and employs only thirty to fifty people.[1] This scale makes data centers a useful setting for studying the local incidence of capital-intensive digital infrastructure. The prior empirical literature is thin; the most rigorous existing study [Hicks, 2024] finds no measurable employment effect of Texas data centers.

[1] Estimates from the IM3 Data Center Atlas (Pacific Northwest National Laboratory) and Synergy Research Group.

At the same time, at least 35 states offer tax incentives targeted specifically at data center construction [Cardinal News, 2026], individual deal packages routinely exceed $100 million per facility, and cumulative subsidies are nearly $20 billion [Good Jobs First, 2024]. These stakes make the policy relevance immediate. Our empirical focus, however, is narrower: the local labor-market response to first entry.

Identification in this setting is not straightforward because data center locations are not randomly assigned. Operators screen heavily on a small set of binding engineering constraints, principally access to high-voltage electrical power and fiber backbone infrastructure, and only after passing those screens compete on costs, incentives, and local negotiations. The set of technically feasible counties is narrow, the firms doing the choosing are sophisticated, and the relevant cost shifters are difficult to observe directly. Cross-county comparisons of treated and untreated counties therefore conflate the causal effect of data center entry with the structural advantages that put a county in the feasible set in the first place.

This paper uses the siting problem as the basis for identification. We construct a novel facility-level dataset of approximately 770 dated US data center openings, covering 93 counties that receive their first large data center between 2008 and 2024, and classify each facility as *hyperscale* (built by cloud and AI operators such as Amazon, Google, Microsoft, and Meta to run their own workloads) or *colocation* (built by data center landlords such as Equinix and Digital Realty who lease space to remote tenants). We link these facilities to county-year employment and wage data from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW) covering 2003 to 2024, and we estimate the effect of first entry on total private employment as well as employment in construction, information, and professional services using a synthetic control (SCM) design. We supplement this approach with IV and staggered-DiD designs.

Our empirical approach is a synthetic control design that compares each treated county to a weighted combination of never-treated counties matched on its pre-treatment employment trajectory [Abadie et al., 2010]. SCM is well suited to the data center setting because treated counties differ from the average never-treated county on level employment, wages, and pre-trend growth, and because staggered treatment timing makes two-way fixed effects unreliable when treatment effects vary across cohorts. Matching on the outcome trajectory at the unit level absorbs both problems directly. We report supportive evidence from a shift-share IV that instruments cumulative installed megawatts with the interaction of a county's pre-existing 345 kV substation count and the leave-one-state-out national data center rollout, along with stacked DiD and Callaway–Sant'Anna estimators. We then write down a simple framework in which a single spillover parameter λ governs how much of the host-county employment effect is reallocation from neighboring counties, and we use contiguous-county

and commuting-zone estimates to bound λ.

Three sets of results follow. First, the synthetic control estimates indicate that data center entry raises total private employment by 4 to 5 percent, construction employment by 11 percent, and information-sector employment by 22 percent at $t = 6$. In level terms at the median treated county, these correspond to roughly 4,400 additional private-sector jobs, 700 additional construction jobs, and 500 additional information-sector jobs six years after first entry. The effects build gradually from $t = 0$, with no evidence of anticipation, and are similar across heterogeneity-robust estimators; the supportive shift-share IV reported in the robustness section points in the same direction.

Second, the information-sector effect is concentrated in counties hosting hyperscale facilities. A triple-difference specification delivers a hyperscale–colocation differential of 31 percentage points in information-sector employment, and a dose-response analysis reinforces this: information-sector agglomeration requires a *cluster* of facilities to materialize, with counties hosting four or more data centers showing significantly larger gains than single-facility counties. The unconditional contrast is even larger, with hyperscale-only counties showing a 43 percent information-sector gain and colocation counties showing no effect. However, this comparison is partly confounded with county size: hyperscale counties have an order-of-magnitude smaller pre-existing information sector, so a given absolute number of new jobs translates into a larger percentage. The triple-difference and size-matched wage results, which hold pre-treatment size fixed, isolate the hyperscale-specific component and remain large. Taken together, these results are consistent with hyperscale operators drawing in concrete local suppliers such as fiber contractors, managed service providers, and network operations centers that do not follow colocation entry. Construction employment, by contrast, rises by a similar amount in both facility types, reflecting the physical build-out of the campuses themselves.

Third, wages rise alongside employment. Average weekly wages in treated counties rise by approximately 3 percent. Using the Quarterly Workforce Indicators from the Census Bureau's LEHD program, we decompose this wage effect into within-job and compositional channels: incumbent workers see earnings rise by 3.3 percent, while new hires earn 3.8 percent more than new hires in synthetic control counties. That both groups benefit indicates genuine local labor market tightening, not compositional reshuffling. FHFA house prices show no statistically significant response, in contrast to the fracking literature [Bartik et al., 2019], though the point estimate is a modest positive drift of 2.0 percent and the FHFA index is known to understate rent capitalization in rural counties; we therefore read the housing evidence as suggesting that capitalization is small relative to wage gains, not that it is zero.

The spillover framework allows us to interpret these magnitudes. Contiguous non-host

counties show small and weakly *positive* spillover effects, the opposite sign from a reallocation story, and commuting-zone aggregations yield a positive total employment effect of 1.7 percent. Together these moments pin λ near zero and imply that the host-county estimates are unlikely to be inflated by within-CZ reallocation. At the local labor-market scale, the evidence is therefore more consistent with net job creation than with pure displacement.

These findings contribute to three literatures. The first is the local labor market multiplier literature initiated by Moretti [2010]: our total employment multiplier is smaller than Moretti's estimates for other technology investments, consistent with data centers representing an extreme point on the capital-to-labor ratio spectrum, but the information-sector effect indicates that even highly capital-intensive investment can generate meaningful technology-sector agglomeration when the investing firm anchors a local supply chain. The second is the place-based employment shocks literature [Kline and Moretti, 2014, Greenstone et al., 2010, Bartik et al., 2019, Slattery and Zidar, 2020]: our setting offers a clean case in which the cost shifter (power infrastructure) is observable and pre-determined, allowing us to ground identification in a known siting margin rather than appealing to general selection-on-observables arguments. The third is the small but growing literature on data centers themselves [Hicks, 2024, Fang and Greenstein, 2025], where the hyperscale-colocation distinction we document is, to our knowledge, one of the first systematic decompositions of how operator type shapes local labor demand. Methodologically, the paper demonstrates that OpenStreetMap edit histories can be used to date the construction of large physical structures, a technique that may be useful for other settings where official permitting data are sparse.

Section 2 reviews the related literature. Section 3 describes the data. Section 4 lays out the institutional setting, describes the synthetic control design, and presents a simple framework for bounding local spillovers. Section 5 presents the main SCM results. Section 6 examines heterogeneity by facility type. Section 7 directly tests for local displacement and applies the spillover bound. Section 9 presents additional robustness checks. Section 10 concludes.

2 Related Literature

A large literature evaluates whether geographically targeted government interventions generate net local employment gains [see Neumark and Simpson, 2015, for a survey]. Kline and Moretti [2014] find that the Tennessee Valley Authority produced large and persistent effects on manufacturing employment. Greenstone et al. [2010] use a "Million Dollar Plants" design and find significant TFP increases among incumbent plants. Busso et al. [2013] evaluate federal empowerment zones and find substantial employment and wage gains. More broadly,

Bartik [2020] synthesizes the evidence on business incentives and finds wide variation in cost-effectiveness, while Slattery and Zidar [2020] document that state tax incentives increasingly flow to large firms and are poorly targeted at distressed areas. Suárez Serrato and Zidar [2016] show that the incidence of state corporate tax cuts falls largely on firm owners rather than workers, raising questions about whether place-based incentives achieve their stated employment goals.

The closest analogue to our setting is Bartik et al. [2019], who study the local economic consequences of hydraulic fracturing, another capital-intensive investment that arrives in rural counties with specific geological endowments. They find employment and income gains but also increased housing costs and health externalities. Our data center setting differs in that the investment is driven by power and fiber infrastructure rather than geology, and the externalities (electricity and water consumption) are less well-documented.

Moretti [2010] estimates a local multiplier of approximately 1.6 non-traded jobs per additional traded-sector job, with larger values for high-technology industries (Moretti and Thulin, 2013 report around 2.5). Section 8 translates our SCM estimates into an implied multiplier of roughly 2.3 at a hyperscale median county under midpoint assumptions about direct on-site employment, placing data centers between Moretti's manufacturing and high-tech benchmarks despite their extreme capital-to-labor ratio. On the digital infrastructure side, Forman et al. [2012] find that the internet raised wages in already-large cities but not elsewhere, while Hjort and Poulsen [2019] document employment gains from fast internet in Africa. Our finding that information sector agglomeration requires a cluster of facilities is consistent with this literature's emphasis on threshold effects in digital infrastructure.

Empirical evidence on the local economic effects of data centers remains thin. The most rigorous existing study is Hicks [2024], who examines data center construction in Texas using county-level employment data and finds no measurable effect on total employment, with a possible exception for professional and technical services near very large facilities. Policy reports underscore the stakes on both sides of the debate: Brookings Institution [2025] emphasizes strategies for converting data center investment into broader local prosperity, while Ohio River Valley Institute [2025] questions whether data centers justify their incentive costs given their low direct employment. We advance this literature in three ways: first, we assemble a national dataset of approximately 770 dated facilities, far larger than any prior study; second, we apply synthetic control methods that more directly address the selection concerns inherent in TWFE designs; and third, we decompose effects by facility type (hyperscale versus colocation), showing that information-sector agglomeration is concentrated in hyperscale investment.

3 Data

3.1 Data Center Locations: The IM3 Atlas

The foundation of the facility-level dataset is the IM3 Open Source Data Center Atlas, developed by the Pacific Northwest National Laboratory (PNNL) as part of the Department of Energy's Integrated Multisector, Multiscale Modeling (IM3) initiative. The atlas identifies 1,242 data center facilities across 46 US states, geocoded to county-level FIPS codes, with information on operator name, facility square footage, and geographic coordinates. The atlas is derived from OpenStreetMap building footprints tagged as data centers, supplemented with facility-level information from operator websites and industry databases.

Table 1 summarizes the atlas. The facilities range from small colocation sites under 10,000 square feet to hyperscale campuses exceeding 3 million square feet. The six largest operators, Amazon Web Services (176 facilities), Digital Realty (70), Meta (61), Google (60), Microsoft (53), and Equinix (38), account for 37 percent of all facilities and a substantially larger share of total floor space. The atlas is dominated by Northern Virginia, which hosts 292 facilities (24 percent of the total); as discussed below, these established hubs are excluded from our analysis sample.

3.2 Opening Dates

The IM3 atlas does not include facility opening dates. We construct opening dates from multiple sources, covering 713 of the 1,242 atlas facilities and an additional 58 non-atlas hyperscale facilities. In the current analysis freeze, we date 617 of the 708 large facilities ($> 100,000$ sqft).

OpenStreetMap Edit Histories. The primary source of opening dates exploits the fact that the IM3 atlas facility identifiers correspond to OpenStreetMap (OSM) way identifiers. We query the OSM API for the edit history of each facility and record the timestamp when the building footprint was first added to the map. For facilities constructed after approximately 2014, when OSM coverage of commercial buildings became comprehensive, the first-mapped date typically falls within 0–2 years of actual construction completion. For earlier facilities, the lag may be larger. This approach yields approximate construction dates for 553 atlas facilities.

State Incentive Records. Several states that offer data center tax incentives maintain public registries of qualifying facilities. The most comprehensive is the Texas Comptroller's

Table 1: Summary Statistics: IM3 Data Center Atlas

	All facilities	Large ($> 100Ksqft$)
Number of facilities	1,242	708
Number of counties	209	111
Square footage:		
Mean	474,256	741,408
Median	132,388	211,010
Top operators:		
Amazon Web Services	176	147
Digital Realty	70	45
Meta	61	61
Google	60	57
Microsoft	53	48
Equinix	38	17
Top states:		
Virginia	292	219
Oregon	94	64
Texas	92	57

registry, which lists 132 qualifying data centers with effective dates, operator names, and occupant information. We also obtain facility-level records from the Nevada Governor's Office of Economic Development and the Ohio Tax Credit Authority.

News and Trade Press. We supplement the above sources with facility opening dates identified from Data Center Dynamics, Data Center Knowledge, company press releases, and state economic development announcements. This approach is most productive for the hyperscale operators (Meta, Google, Microsoft), whose facility openings are widely covered. We also conduct a targeted search for AWS facilities, whose parent company Amazon operates through subsidiary entities (principally Vadata, Inc.) that make individual facility identification difficult.

Date Validation. We validate the OSM-derived dates against the Texas Comptroller's registry of qualifying data centers, which records the effective date of tax exemption eligibility for 123 facilities. Matching 11 facilities that appear in both the Comptroller registry and the IM3 atlas, we find a median lag of zero years (mean 0.8 years) between OSM and Comptroller dates, with 82 percent of matches falling within ± 2 years. The two largest discrepancies (LinkedIn Richardson, 4 years; State Farm Richardson, 3 years) involve pre-2015 facilities, consistent with sparser OSM coverage of commercial buildings before 2014.

Because the OSM-derived dates may lag actual facility openings by up to two years, we further assess sensitivity to date measurement error by re-estimating the SCM with treatment timing shifted by -1 and -2 years. The total employment effect is stable across shifts (4.2, 4.2, and 3.5 percent), and the information sector effect varies from 17 to 29 percent (Table 17 in Section 9).

3.3 Employment Outcomes: QCEW

County-year employment and wage data come from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW). The QCEW is based on administrative records from the unemployment insurance system and covers approximately 95 percent of US employment. We obtain annual average employment levels and weekly wages for four industry categories at the county level: total private employment (NAICS 10), construction (NAICS 23), information (NAICS 51), and professional and technical services (NAICS 54).[2] The panel covers 2003–2024.

[2]The 2022 NAICS revision expanded NAICS 518 to explicitly include cloud computing and data center operators. Because our analysis uses 2-digit NAICS 51, this reclassification does not affect our series.

3.4 Analysis Sample

The analysis sample in the current paper consists of 93 treated counties that receive their first large data center (defined as a facility exceeding 100,000 square feet) between 2008 and 2024, and approximately 3,040 control counties that do not host a large data center during the sample period. We exclude counties with large data centers opening before 2008 to ensure at least five years of pre-treatment data, and exclude 17 control counties that host large undated facilities (potential control contamination). The resulting panel contains approximately 69,000 county-year observations for total private employment, with somewhat fewer observations for sector-specific outcomes due to BLS disclosure suppression in small counties.

Of the 93 treated counties, 33 host only hyperscale facilities, 23 host only colocation facilities, 12 are "mixed" (hosting both types), and 25 host facilities by smaller or unclassified operators. This variation in facility type enables us to test whether the employment effects of data center construction depend on the type of operator, a distinction with direct policy relevance since state incentive programs do not typically differentiate between hyperscale and colocation investment.

Our research design estimates the effect of *new data center entry* into previously untreated counties. By construction, this excludes established data center hubs that hosted large facilities before 2008, including Northern Virginia (292 atlas facilities), the Chicago metro area, and parts of the Dallas-Fort Worth metroplex. These hubs account for a substantial share of total US data center employment, but their selection into early adoption reflects deep structural advantages (proximity to internet exchange points, fiber backbone infrastructure, federal agencies) that make them poor candidates for causal inference. The 93 treated counties in the current analysis sample represent the policy-relevant margin: locations where state and local incentives are actively deployed to attract new facilities. Our main estimates should therefore be interpreted as the effect of *entering* the data center market. Because facility commissioning is often staggered and opening dates are measured at annual frequency, the treatment year should be read as the year of first large-facility entry into the county rather than the exact month the site reached full operation. Because roughly half of treated counties subsequently attract additional facilities, the estimand captures the full trajectory following first entry, including any endogenous accumulation of follow-on investment. In Section 9, we decompose this effect by facility count using a dose-response design, showing that total employment gains are driven by first entry while information sector agglomeration requires a cluster.

4 Institutional Setting and Identification

4.1 The Data Center Location Problem

Data center sites are selected from a narrow set of technically feasible locations. Operators screen first on four binding constraints: high-capacity electrical power, access to fiber-optic backbone, land with permitting pathways that admit heavy industrial use, and, for water-cooled campuses, access to municipal or surface water at volume. Only after these engineering screens are passed do operators compete on costs, tax incentives, and local negotiations. This pattern is consistent with the broader location-choice literature [Alcácer and Chung, 2014, Alcácer and Delgado, 2016] and with recent evidence that colocation facilities value proximity to dense network ecosystems while cloud providers are more willing to trade proximity for cheaper land and scalable power access [Fang and Greenstein, 2025].

Of these screens, power is first-order. A modern hyperscale campus requires 100 to 500 megawatts of firm, continuous electrical load, an order of magnitude larger than a typical manufacturing facility. Delivering that load requires proximity to high-voltage transmission infrastructure, typically 345 kV or above, that can source from multiple generation points and absorb the demand without destabilizing local distribution. Counties with existing high-voltage substation infrastructure therefore enter the feasible set at low marginal cost, while counties without it can only enter after multi-year utility planning and substation construction.

Public interconnection data underscore that power is the binding siting margin, even though they do not provide a clean national panel of data center load requests. NYISO, the one queue in our assembled data that separately records large-load applications, contains 30 load requests through 2024, seven with names explicitly referring to data centers or digital campuses (Table 2). These requests are tied to specific substations or transmission lines, highlighting that treated counties come from a much narrower feasible set than the full US county distribution.

This siting logic has two implications for how we read cross-county variation in data center entry. First, the comparison group must approximate the small set of counties with power and fiber conditions that could plausibly host a large facility; counties outside that set are not informative counterfactuals. Second, even within the feasible set, treated counties differ from never-treated counties on observable pre-trends, so an effective design must match on the pre-treatment outcome trajectory itself rather than rely on parallel-trends conditional on fixed effects. These two requirements motivate the synthetic control design we describe next.

Table 2: Illustrative Named Data Center Load Requests in the NYISO Queue

Project	County	MW	Queue date	Status
North Country Data Center	St. Lawrence	176	2020-01-22	Active
Niagara Digital Campus	Niagara	140	2024-04-09	Active
Cayuga Compute	Tompkins	88	2024-04-24	Active
Hudson Valley Data Center	Rockland	50	2024-07-02	Active
450 Broadway Datacenters (Buchanan)	Westchester	200	2024-08-07	Active
Greenidge 200 MW Data Center Project	Yates	200	2024-12-20	Active

Notes: Illustrative requests from the raw NYISO interconnection queue with project names containing "data center," "digital," or "compute" and queue dates through 2024. MW values and queue status are reported directly from the queue. NYISO is unusually transparent about large-load requests; the other regional queue files assembled for this project are overwhelmingly generator-oriented and generally do not separately identify data center load applications.

4.2 Synthetic Control

Our primary identification strategy is a synthetic control design. SCM constructs, for each treated county, a weighted combination of never-treated counties that matches its pre-treatment employment trajectory [Abadie et al., 2010, Abadie, 2021]. Weights are non-negative and sum to one, with no intercept, ensuring the counterfactual is a convex combination of real counties. The donor pool consists of the 200 nearest never-treated counties by pre-treatment employment size, and we exclude treated counties with pre-treatment RMSE above 0.15 log points.

SCM is well suited to the data center setting for two reasons. First, treated counties differ from the average never-treated county on level employment, wages, and pre-trend growth (Table 3); a design that matches on the outcome trajectory itself absorbs these differences directly rather than asking fixed effects to do the work. Second, treatment timing is staggered across counties, and TWFE estimators with staggered timing can produce biased estimates when treatment effects are heterogeneous across cohorts [Goodman-Bacon, 2021, de Chaisemartin and D'Haultfœuille, 2020]. SCM sidesteps both concerns by matching at the unit level and aggregating only after each treated county has its own counterfactual. Section 9 reports a complementary shift-share IV based on pre-existing high-voltage substation density, along with stacked DiD and Callaway–Sant'Anna estimators; we treat these as supportive checks rather than as coequal designs.

4.3 A Framework for Bounding Spillovers

The SCM design recovers the effect of data center entry on a *host* county relative to non-host donor counties. This comparison can be contaminated if data centers displace economic activity from donor counties into host counties, a SUTVA violation that no reduced-form design can eliminate by construction. We write down a simple framework that uses our CZ-level and contiguous-county estimates to bound the true local effect, and we then check that the two estimates are internally consistent.

Let τ denote the true causal effect of data center entry on log private employment in the host county, and let τ_N denote the effect on the *ring* (non-host counties in the same commuting zone). The host-county SCM estimator $\hat{\tau}_H$ recovers

$$\hat{\tau}_H \;=\; \tau \;-\; w_N\,\tau_N, \tag{1}$$

where w_N is the share of the synthetic control donor pool drawn from ring counties. In our design donors are selected on pre-treatment employment size rather than geography, and re-running SCM with all same-CZ donors excluded leaves the point estimates within 0.2 percentage points of the baseline, so $w_N \approx 0$ and (1) reduces to $\hat{\tau}_H \approx \tau$.

The CZ-level SCM estimator $\hat{\tau}_{CZ}$ compares total CZ employment to never-treated CZs, and to first order recovers

$$\hat{\tau}_{CZ} \;=\; \phi_H\,\tau \;+\; (1-\phi_H)\,\tau_N, \tag{2}$$

where ϕ_H is the pre-treatment employment share of the host county within its CZ. The median host share across our 92 treated counties is $\phi_H = 0.248$.

Two observations discipline τ_N. The direct contiguous-county SCM yields $\hat{\tau}_N$ of 1.1 to 1.4 percent at $t = 1$ through $t = 3$, fading toward zero by $t = 6$; the point estimates are non-negative at every horizon, so we can impose $\tau_N \geq 0$. Given this sign restriction, equation (2) delivers the upper bound

$$\tau \;\leq\; \frac{\hat{\tau}_{CZ}}{\phi_H}, \tag{3}$$

and equation (1) delivers the lower bound $\tau \geq \hat{\tau}_H$. Combining,

$$\hat{\tau}_H \;\leq\; \tau \;\leq\; \frac{\hat{\tau}_{CZ}}{\phi_H}. \tag{4}$$

Plugging in the headline SCM estimates ($\hat{\tau}_H = 4.0$ percent for total private employment at the median event time) and the CZ-level estimate ($\hat{\tau}_{CZ} = 1.7$ percent) at the median host share gives

$$4.0\% \;\leq\; \tau \;\leq\; 6.8\%.$$

The interval is non-empty, which establishes that the two estimates are internally consistent with a non-negative ring spillover. Solving equations (1) and (2) jointly for τ_N under $w_N \approx 0$ yields the implied ring effect $\hat{\tau}_N^{\text{implied}} = (\hat{\tau}_{CZ} - \phi_H \hat{\tau}_H)/(1 - \phi_H) \approx 0.9$ percent, which is within the 1.1 to 1.4 percent range obtained directly from the contiguous-county SCM. The framework therefore recovers the ring effect out-of-sample from the host and CZ moments to within their sampling variability, which we read as evidence that the host-county estimates are net of within-CZ reallocation rather than inflated by it.

A formal spillover parameter with propagated standard errors. To move beyond the interval bound, we define the spillover parameter

$$\lambda \equiv \tau_N / \tau, \tag{5}$$

the ring effect expressed as a fraction of the host effect. A value of $\lambda = 0$ means ring counties are unaffected; $\lambda < 0$ means reallocation; $\lambda > 0$ means positive spillover to the ring. Substituting (5) into (2) and solving under $w_N \approx 0$ gives the point estimate

$$\hat{\lambda} = \frac{\hat{\tau}_{CZ} - \phi_H \hat{\tau}_H}{(1 - \phi_H)\hat{\tau}_H}. \tag{6}$$

Plugging in $\hat{\tau}_H = 0.0408$ (total-employment SCM at $t = 6$), $\hat{\tau}_{CZ} = 0.0173$, and $\phi_H = 0.248$ yields $\hat{\lambda} = 0.234$. The delta-method standard error, taking $\hat{\tau}_H$ and $\hat{\tau}_{CZ}$ as independent (they are estimated on non-overlapping aggregations of the same panel, and covariance across levels of aggregation is positive, so treating them as independent is conservative on the variance), is

$$\text{SE}(\hat{\lambda}) = \sqrt{\left(\tfrac{1}{(1-\phi_H)\hat{\tau}_H}\right)^2 \text{Var}(\hat{\tau}_{CZ}) + \left(\tfrac{\hat{\tau}_{CZ}}{(1-\phi_H)\hat{\tau}_H^2}\right)^2 \text{Var}(\hat{\tau}_H)}.$$

With $\text{SE}(\hat{\tau}_H) = 0.0101$ (the standard error of the mean host-county gap at $t = 6$ across the 47 treated units that reach the six-year horizon, computed as the cross-unit standard deviation divided by $\sqrt{47}$) and $\text{SE}(\hat{\tau}_{CZ}) = 0.0093$ (from the CZ-level SCM permutation distribution), this delivers $\text{SE}(\hat{\lambda}) = 0.33$ and a 95 percent confidence interval of $[-0.42, 0.89]$. We read this interval conservatively. The load-bearing number is the lower endpoint: the data reject any reallocation story in which ring counties lose more than about forty cents of employment per dollar gained by the host, which is the margin that would overturn the welfare interpretation of the host-county headline. The point estimate itself is positive, consistent with the direct contiguous-county SCM and with the framework's out-of-sample prediction $\hat{\lambda} \cdot \hat{\tau}_H = 0.96$ percent falling inside the 1.1–1.4 percent range recovered from contiguous counties directly, but the confidence interval contains zero and we do not claim to have pinned down a statistically

significant amplification. Two caveats apply. First, the width of the CI is driven by SE($\hat{\tau}_{CZ}$) rather than SE($\hat{\tau}_H$), so the binding constraint on sharpening λ is the modest CZ-level sample ($n = 58$ treated CZs), not the host-county design. Second, the framework bounds *within*-CZ reallocation only: if data centers displace employment across CZ boundaries, $\hat{\tau}_{CZ}$ already absorbs that leakage and $\hat{\lambda}$ understates displacement. The hydraulic-fracturing literature [Bartik et al., 2019] faces the same boundary-of-validity issue, and we read the exercise as ruling out the within-CZ reallocation channel that is most directly testable in our data, not as a claim about the total geographic footprint of the shock.

The framework is deliberately spare. It imposes a two-region partition of the commuting zone, treats the ring effect as a single scalar rather than a spatial decay function, and does not model general-equilibrium effects on prices and wages outside the local labor market. A full quantitative spatial equilibrium treatment is left for future work. Within the terms set out here, however, the host-county SCM estimate $\hat{\tau}_H$ is a lower bound on τ, the CZ-level estimate $\hat{\tau}_{CZ}$ divided by the host employment share is an upper bound, and the two moments yield an interval for the true local effect that is consistent with the contiguous-county evidence.

4.4 Threats to Identification

The main residual threat to the SCM design is selection on unobserved trends: conditional on the engineering and infrastructure screens described in Section 4.1, data centers may still locate in counties that are already growing for reasons we do not observe. Pre-treatment balance statistics (Table 3) show that treated counties are larger, have higher wages, and exhibit faster pre-treatment employment growth than the average control county. We address this threat in three complementary ways. First, the SCM design matches on the pre-treatment employment trajectory itself, and the event study (Figure 1) provides a visual test for differential pre-trends. Second, we match treated counties to control counties within the same pre-treatment employment size quintile. Third, we present results separately for rural and urban counties and for hyperscale and colocation facilities, to assess whether the effects are driven by specific subgroups.

Anticipation and operator secrecy. A distinctive feature of data center site selection is the use of shell companies and non-disclosure agreements to conceal the operator's identity during the planning and construction phases. Amazon operates through Vadata, Inc.; Google uses subsidiaries such as Design LLC and Sharka LLC; and Virginia FOIA investigations have documented NDAs between localities and data center operators that suppress public information about incoming facilities. This secrecy limits the ability of local labor markets to anticipate the treatment: while a large construction project is visible to the community,

the identity of the operator and the nature of the facility (hyperscale vs. colocation) are often unknown until after construction is complete. The gradual build-up of the information sector effect from $t = 0$ onward, rather than a pre-treatment jump, is consistent with this lack of anticipation: local IT firms and managed service providers cannot pre-position around a facility whose operator they do not yet know.

Table 3: Pre-Treatment Balance: Treated vs. Control Counties

	Treated	Control	Difference
Log pre-treatment employment	11.58	9.49	2.09
Pre-treatment employment growth (2003–07)	13.6%	7.8%	5.8 pp
Average weekly wage ($)	748	576	172
Number of counties	92	2,948	

Notes: Pre-treatment characteristics computed as county-level means over 2003–2007. Employment growth is the percentage change in total private employment from 2003 to 2007. One treated county with fewer than 3 pre-period observations is excluded from this table; the full analysis sample contains 93 treated counties.

5 Results

5.1 Main Results

A standard TWFE regression of log employment on a post-treatment indicator with county and year fixed effects yields large and significant effects across all sectors: 15 percent for total private employment, 18 percent for construction, 41 percent for information, and 29 percent for professional services (Table 19 in Section 9). However, as documented in Section 4, treated counties were growing faster than controls before data center construction, inflating the TWFE estimates. We therefore turn to the synthetic control method as our preferred specification.

Applying the SCM procedure described in Section 4 yields 90 well-matched treated counties for total employment (the per-panel sample sizes in Figure 1 reflect the subset with observations at each event time). Figure 1 presents the results. All four panels show flat pre-treatment gaps, confirming that the synthetic controls match the treated counties' pre-treatment trajectories. Post-treatment, the gaps are positive and growing in all sectors.

Total private employment shows a gradual increase reaching 4.2 percent by $t = 6$ ($p < 0.01$), comparable to the 1.5 percent TFP gain that Greenstone et al. [2010] document for "Million Dollar Plant" counties. Alternative estimators that are robust to heterogeneous treatment effects in staggered designs (Callaway–Sant'Anna, stacked DiD) yield similar estimates of 5 percent (Section 9). Construction employment jumps immediately at $t = 0$ and grows to 10.9 percent by $t = 6$ ($p < 0.01$), reflecting both the initial construction workforce and ongoing facility expansion. Information sector employment builds gradually to 22.4 percent by $t = 6$ ($p < 0.05$), consistent with agglomeration of related IT activity around the data center. To assess whether this effect is mechanical (reflecting data center employees classified under NAICS 51), we examine the 3-digit NAICS composition: at the median treated county, NAICS 518 (Data Processing, Hosting, and Related Services) accounts for only 14 percent of total NAICS 51 employment, with telecommunications (NAICS 517) and other subsectors comprising the remainder. Even a large increase in NAICS 518 employment can therefore explain only a small share of the overall information sector effect.[3] Professional services employment grows to 16.8 percent by $t = 6$ ($p < 0.05$).

To assess the plausibility of these magnitudes, we decompose the SCM estimates by the number of large facilities in each treated county. Of the 93 treated counties, 46 host exactly one large data center, while 47 host two or more (median 2, mean 4.9). Single-facility counties show a significant 5.8 percent total employment effect ($p < 0.01$) and 13.0 percent construction effect ($p < 0.01$), but no significant information sector effect (3.2 percent, $p > 0.10$). Multi-facility counties, by contrast, drive the information sector result (25.8 percent, $p < 0.01$) but show a more modest total employment effect (2.5 percent, $p < 0.10$). This pattern suggests that a single data center generates construction activity and modest total employment gains, but the information sector agglomeration effect, the attraction of related IT services firms, managed service providers, and network operators, requires a *cluster* of facilities to materialize. The pooled SCM estimates should therefore be interpreted as the effect of entering the data center market, which typically involves the accumulation of multiple facilities over time, rather than the effect of a single facility opening in isolation.

5.2 Wage Effects

Figure 2 decomposes the wage effect by sector using the same SCM methodology. Average weekly wages across all private sectors rise by 2.8 percent by $t = 6$ ($p < 0.01$), with effects beginning immediately at $t = 0$. Information sector wages show the largest effect, consistent

[3] NAICS 518 data are available from the QCEW for 59 of 93 treated counties; the remainder are suppressed by BLS for confidentiality. Among counties with available data, NAICS 518 ranges from 1 to 78 percent of NAICS 51 (mean 17 percent).

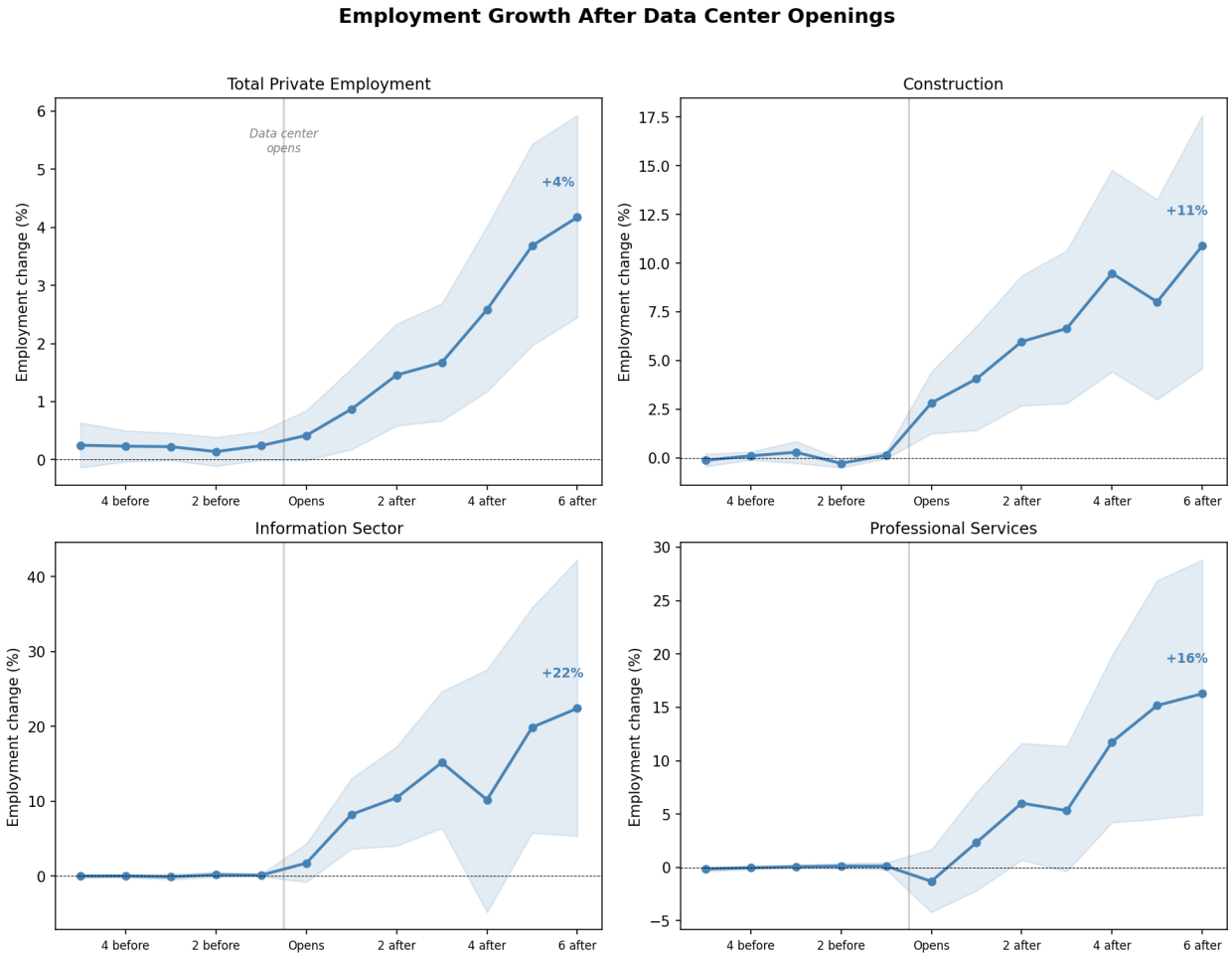

Figure 1: Synthetic Control Event Study: Employment Around Data Center Openings

Notes: Each panel plots the average gap between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. Synthetic controls are constrained convex combinations of the 200 nearest donor counties by pre-treatment employment. Counties with pre-treatment RMSE > 0.15 are excluded. The three sector panels share a common y-axis. Event time 0 corresponds to the year of first large data center opening. Formal inference using the prediction interval approach of Cattaneo et al. [2021] is reported in Table 14 and Section 9.

with data centers attracting high-paying technology jobs. Construction and professional services wages also rise, reflecting increased demand for building trades and skilled services. The wage effects are uniformly larger than the corresponding employment effects, suggesting that data centers raise local wages through a combination of direct high-wage job creation and tightening of local labor markets. To decompose the wage effect into compositional and within-job channels, we use the Quarterly Workforce Indicators (QWI) from the Census Bureau's LEHD program, which reports average monthly earnings separately for *stable* (full-quarter, continuing) workers and *new hires*. The QWI data cover all private-sector employment at the county-quarter level. Cell suppression is minimal (fewer than 0.2 percent of observations for both treated and control counties). Because the QWI reports earnings at the all-industry level, we cannot isolate within-occupation wage growth; the continuing-worker effect captures within-establishment wage growth, which could reflect promotions or role changes within firms as well as direct wage increases.

Applying the SCM methodology to QWI earnings, we find that both channels contribute to the wage gains. Continuing workers see earnings rise by 3.3 percent by $t = 6$ ($p < 0.05$), while new hires see a similar effect (3.8 percent, $p < 0.01$). These magnitudes are consistent with the QCEW wage estimate of 2.8 percent and economically plausible. That both continuing workers and new entrants benefit indicates that data centers tighten local labor markets broadly, not merely through compositional shifts from high-wage firms entering the county.

6 Heterogeneity

We examine heterogeneity along four dimensions: facility type, facility size, county type, and treatment timing.

6.1 Heterogeneity by Facility Type

The most policy-relevant source of heterogeneity is *facility type*: do hyperscale data centers (operated by cloud and AI companies) generate different local employment effects than colocation facilities (operated by wholesale data center providers)? This distinction matters because hyperscale operators bring their own ecosystem of technology suppliers and managed service providers, while colocation providers primarily offer space and power to remote third-party tenants.

Hyperscale and colocation counties differ substantially in pre-treatment characteristics. Hyperscale counties are much smaller (median private employment of 36,000 versus 366,000

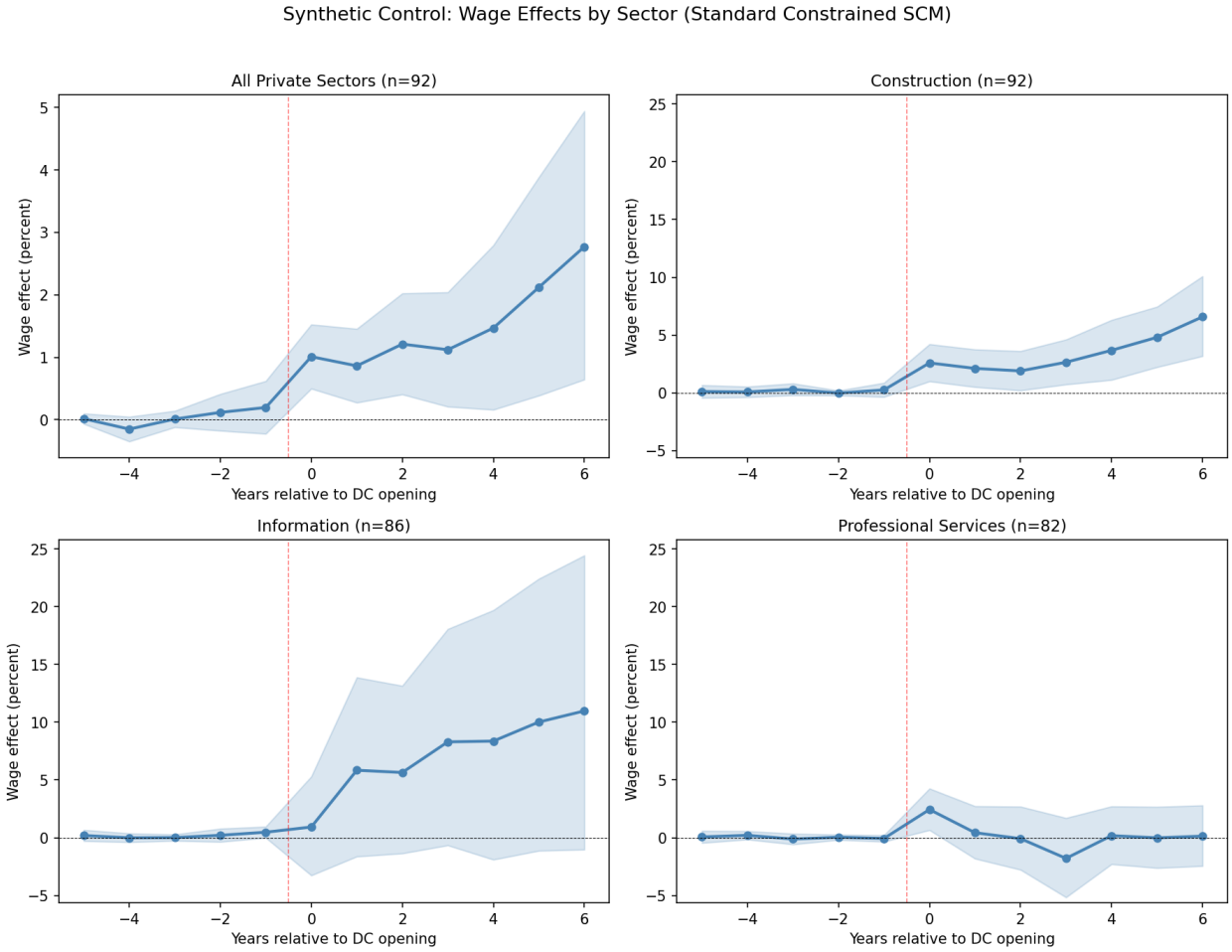

Figure 2: Synthetic Control: Wage Effects by Sector

Notes: Each panel plots the average gap in log average weekly wages between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. Methodology as in Figure 1. The three sector panels share a common y-axis.

for colocation counties), have lower wages ($628 versus $896 weekly), and have smaller information sector employment shares (1.6 versus 3.5 percent). These differences reflect the location strategies of the two operator types: hyperscale facilities seek cheap land and power in less populated areas, while colocation providers locate near existing network infrastructure in major metros. Reassuringly, pre-treatment employment *growth rates* are balanced across types (12.2 versus 11.1 percent, $p = 0.79$), supporting the parallel trends assumption within each group.

Table 5 and Figure 3 present the SCM results by facility type. All types show positive total employment effects, but the information sector diverges sharply: hyperscale-only counties show a 43 percent increase ($n = 25$), while colocation counties show a -5 percent decline ($n = 20$). Mixed counties fall in between (8 percent, $n = 11$). A triple-difference specification quantifies this differential: interacting the post-treatment indicator with a hyperscale county indicator yields a conditional difference of 31 percentage points ($p < 0.05$) for information sector employment.

Because hyperscale and colocation counties differ on pre-treatment size, we verify that the type divergence is not driven by county size itself. Table 4 restricts the comparison two ways. Panel A reports SCM effects on the common-support sample, retaining only counties with pre-treatment employment inside the overlap of the two type distributions (26 of 33 hyperscale and 18 of 22 colocation counties). Panel B reports SCM effects on size-caliper matched pairs, in which each colocation county is paired with the nearest hyperscale county in log pre-treatment employment, without replacement. Both restrictions preserve the qualitative pattern. On the common-support sample, the information-sector gap is +38 percent in hyperscale counties and -4 percent in colocation counties; on the matched-pair sample, the difference is +36 percentage points with $t = 5.0$. Size matching therefore tightens rather than erases the hyperscale–colocation divergence, and the base-rate concern, that hyperscale gains look large because they start from a small information sector, does not survive restriction to counties whose information base is comparable to the colocation group.

This pattern implies that hyperscale operators generate substantially more local IT agglomeration than colocation providers. The most concrete interpretation is that hyperscale facilities draw in local suppliers such as network operations centers, managed service providers, and fiber and power contractors, raising information-sector employment in the host county. The localization of these spillovers at the county level likely reflects supply-chain proximity rather than labor pooling: managed service providers, fiber contractors, and network operations centers need physical access to the campus for equipment installation, maintenance, and low-latency interconnection, creating incentives to locate within a short drive of the facility. This distinguishes data centers from other technology investments where

Table 4: Hyperscale vs. Colocation on Size-Restricted Samples

Total private		Information	
Hyperscale	Colocation	Hyperscale	Colocation
Panel A: Common-support sample			
$+3.2^{***}$	$+1.5^{***}$	$+37.8^{***}$	-3.8
$n = 23$	$n = 17$	$n = 20$	$n = 17$
Panel B: Size-caliper matched pairs			
$+4.4^{***}$	$+1.9^{***}$	$+29.4^{***}$	-4.9
$n = 19$	$n = 20$	$n = 17$	$n = 20$

Notes: Each cell reports the average post-treatment SCM gap (in percent, exponentiated from the log-employment gap) across counties in the indicated sample. Standard errors are computed from the cross-county distribution of average post gaps. Panel A restricts both groups to the common support of pre-treatment total private employment, [14K, 684K]. Panel B pairs each colocation county to its nearest hyperscale county in log pre-treatment employment, without replacement, yielding 22 matched pairs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

agglomeration operates primarily through thick labor markets [Moretti, 2010]. Colocation providers, by contrast, lease space to remote tenants whose operations staff typically remain at the tenant's headquarters rather than at the facility site, generating less demand for local IT employment. The 43 percent information-sector effect in hyperscale counties should be interpreted in light of the small base: at the median hyperscale county, this represents approximately 290 new information-sector jobs, a modest absolute gain that is nonetheless large relative to the county's pre-existing information sector. This base effect does not invalidate the finding, but it does suggest that the percentage effect would be smaller in counties with larger pre-existing information sectors.

A direct check on the anchor-ecosystem interpretation is whether the information-sector workers entering hyperscale counties are higher paid, as predicted by a story in which the entrants are technical staff and on-site engineering rather than generic clerical or sales occupations. Table 6 reports SCM estimates of the effect of entry on log average wages by sector, split by facility type. The information-sector wage rises by 19.0 percent in hyperscale counties (SE = 6.0) and shows no effect in colocation counties (-3.8 percent, SE = 4.3); the hyperscale–colocation difference is 22.8 percentage points ($p = 0.002$). Total private wages, by contrast, rise by a similar modest amount in both types (1.3 versus 2.0 percent)

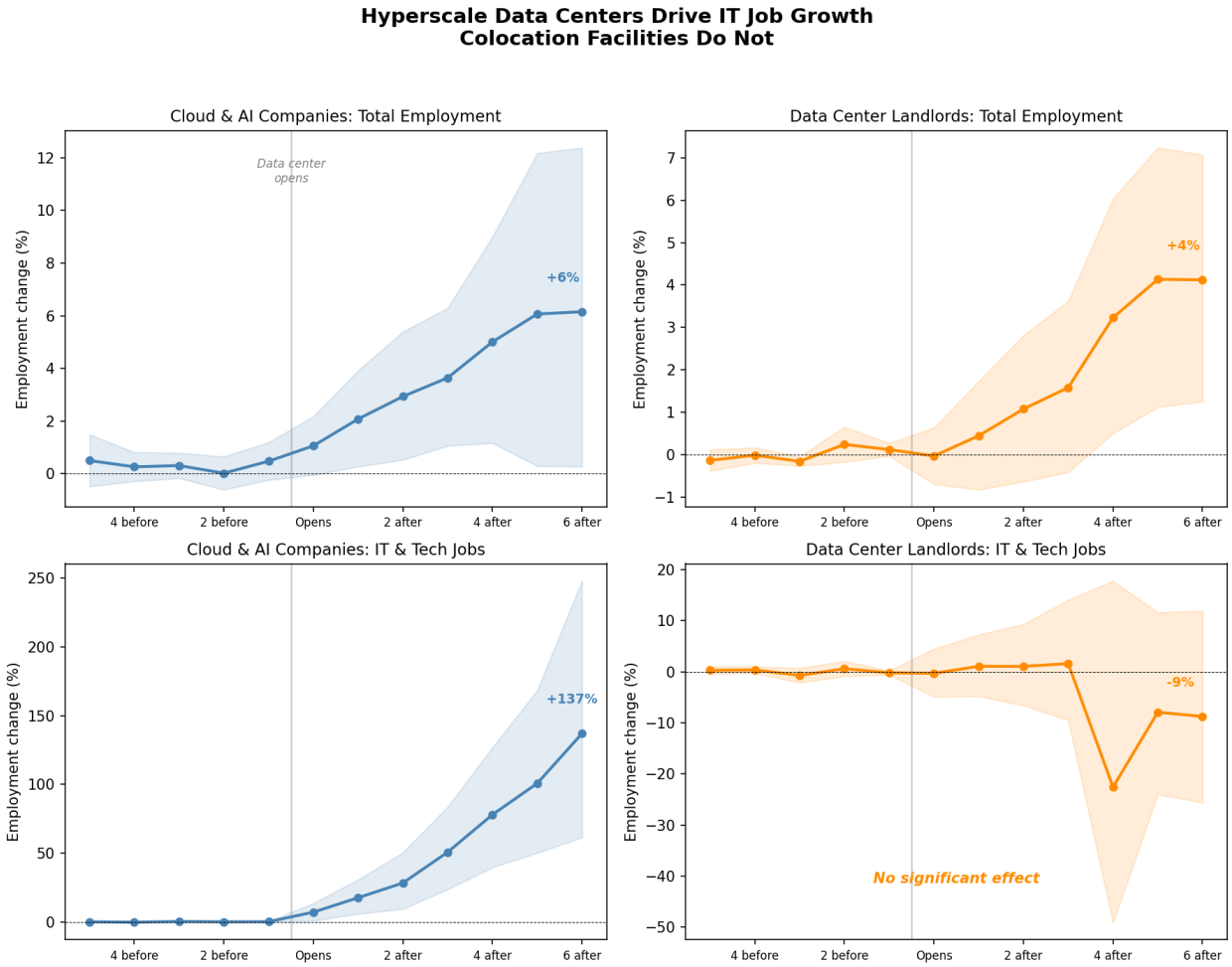

Figure 3: Synthetic Control by Facility Type: Hyperscale vs. Colocation

Notes: Each panel plots the average gap between treated counties and their synthetic controls, with 95 percent confidence intervals. Left column: hyperscale-only counties. Right column: colocation-only counties. Top row: total private employment. Bottom row: information sector employment. Per-panel sample sizes reflect counties with observations at each event time. Methodology as in Figure 1.

Table 5: SCM Employment Effects by Data Center Facility Type

	Total private	Information	N
All DC counties	$+4.2\%^{***}$	$+22.4\%^{**}$	90 / 80
Hyperscale only	$+3.3\%^{***}$	$+43.3\%^{***}$	30 / 25
Colocation only	$+1.9\%^{***}$	-4.9%	20 / 20
Mixed	$+1.1\%^{***}$	$+7.6\%^{**}$	12 / 11

Notes: Each cell reports the average post-treatment SCM gap (in percent) between treated counties and their synthetic controls at $t = +6$. Significance based on cross-county standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. N reports the number of matched treated counties for total private / information sector. Hyperscale counties contain only hyperscale operators. Colocation counties contain only colocation/wholesale operators. Mixed counties contain both types.

and do not differ significantly across types. The same facility-type split that produces a 31-percentage-point employment divergence in the information sector therefore also produces a 23-percentage-point wage divergence in the information sector and essentially none in the aggregate. This is what one would expect if hyperscale entry pulls in higher-paid technical workers concentrated in information services. It is harder to reconcile with a story in which colocation simply fails to attract workers: colocation's zero employment effect is paired with a near-zero wage effect, so the type difference appears to operate through who enters rather than through wage compression at the bottom of the distribution.

Direct test of the anchor-firm mechanism: establishment entry. The wage-premium result is consistent with an anchor-firm story but does not by itself distinguish firm entry from incumbent hiring. A sharper test is whether hyperscale entry raises the *count* of establishments in the information and construction sectors, not merely their headcount. If hyperscale operators are anchoring a local supply chain of fiber contractors, managed service providers, and network operations centers, we should observe new physical establishments opening in the host county, not only existing establishments hiring more workers. If hyperscale entry instead reflects churn within incumbents or reclassification of existing workers, establishment counts should be unchanged. Table 7 applies the same SCM design to county-year log establishment counts from QCEW for NAICS 23 (construction), 51 (information), and 54 (professional services). Hyperscale-only entry raises the count of construction establishments by 18.1 percent at $t = 6$ (SE = 6.9) and the count of information-sector establishments by

Table 6: Wage Effects by Facility Type

Outcome	Hyperscale	Colocation	Hyp − Col
Total private wage	+1.31%**	+1.99%*	−0.68%
	(0.52)	(1.07)	(1.19)
Information wage	+19.03%***	−3.76%	+22.79%***
	(6.03)	(4.32)	(7.42)
Information emp (ref.)	+26.88%***	−4.67%	+31.54%***
	(6.60)	(5.21)	(8.41)

Notes: Each cell reports the average post-treatment SCM gap across treated counties in the indicated group. Standard errors are computed from the cross-county distribution of average post-treatment gaps. The third row re-reports the information-sector employment result from Table 5 for comparison. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

26.2 percent (SE = 12.6), both statistically significant. Professional-services establishment entry is positive but imprecisely estimated (+7.5 percent, SE = 6.8). Colocation-only entry, by contrast, generates no establishment entry in any of the three sectors: construction estabs are +1.5 percent (SE = 2.9), information −0.2 percent (SE = 7.5), and professional services −1.7 percent (SE = 3.2). The information-sector establishment effect in hyperscale counties tracks the 22-percent information-sector employment effect closely, consistent with the employment gain being delivered by new firms entering the host county rather than solely by hiring at incumbents. The absence of establishment entry in colocation counties, paired with their zero employment and wage effects, points in the same direction. At the same time, QCEW establishment counts do not let us observe supplier relationships directly, so we read this as strong supportive evidence for the anchor-firm mechanism rather than a definitive supplier-linkage test.

Dose-response. If hyperscale entry generates local information-sector agglomeration, larger local clusters should produce larger effects. We estimate the Callaway and Sant'Anna [2021] heterogeneity-robust ATT separately for counties grouped by facility count: 1 facility ($n = 45$), 2–3 facilities ($n = 11$), and 4+ facilities ($n = 36$). The results reveal a striking asymmetry. Total private employment effects are flat across dose bins (4–5 percent regardless of facility count), suggesting that the total employment gain is driven primarily by first entry and does not scale much with subsequent investment. Information-sector employment, by contrast, shows a clear dose-response gradient: 7 percent for single-facility counties (not significant), 8 percent for 2–3 facilities, and 23 percent for counties with 4 or

Table 7: SCM Estimates for Establishment Counts, by Facility Type

Sample	Construction (NAICS 23)	Information (NAICS 51)	Professional services (NAICS 54)
All treated	+7.8*** (2.5)	+9.4* (4.9)	+1.8 (2.2)
Hyperscale-only	+18.1*** (6.9)	+26.2** (12.6)	+7.5 (6.8)
Colocation-only	+1.5 (2.9)	−0.2 (7.5)	−1.6 (3.2)
N at $t = 6$ (all / hyp / colo)		45 / 12 / 15	

Notes: SCM estimates at $t = 6$ for log county-year establishment counts from QCEW (annual singlefile, private ownership), for NAICS 23 (construction), 51 (information), and 54 (professional, scientific, and technical services). The same matching procedure used for employment outcomes is applied unchanged: nearest-200 donors by pre-treatment log employment, SLSQP-constrained weights, pre-period RMSE screen. Cross-treated standard errors in parentheses. Hyperscale-only counties host at least one hyperscale facility and no colocation facilities; colocation-only is the reverse. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

more facilities ($p < 0.01$). This confirms that the information-sector agglomeration effect requires a *cluster* of facilities to materialize. A single facility does not generate sufficient local demand to support managed service providers or network operations centers, while a cluster of facilities creates a local market large enough to sustain these upstream suppliers.

6.2 Other Sources of Heterogeneity

Table 8 examines heterogeneity by county characteristics and treatment timing using SCM.

Table 8: SCM Heterogeneity by County Characteristics and Cohort

Subsample	Gap at t=+6	SE	N	Implied effect
By county type:				
Rural (below-median pre-employment)	0.055**	(0.021)	45	5.6%
Urban (above-median pre-employment)	0.032***	(0.010)	45	3.3%
By treatment cohort:				
2008–2014	0.029***	(0.011)	30	2.9%
2015–2019	0.062***	(0.020)	21	6.4%

Notes: Each row reports the average SCM gap at $t = +6$ for total private employment, estimated on the indicated subsample. SE is the cross-county standard error. The 2020–2024 cohort is omitted because insufficient post-treatment years are available for $t = +6$ estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Two patterns emerge. First, *rural counties* show larger effects (5.6 percent, $p < 0.05$) than *urban counties* (3.3 percent, $p < 0.01$), consistent with the hypothesis that a single large facility represents a larger relative shock in less populated areas. Both estimates are statistically significant.

Second, the effects are positive across treatment cohorts. The 2015–2019 cohort shows a larger effect (6.4 percent) than the 2008–2014 cohort (2.9 percent), likely reflecting the rising scale of modern data center facilities.

The sustained construction employment effect (11 percent by $t = 6$) reflects the fact that, unlike a typical factory opening where construction employment spikes and then dissipates, data center campuses are built in phases over many years. Of the 93 treated counties, 47 host multiple large facilities, and many receive additional facilities within 2–4 years of the first. This ongoing construction activity, combined with the ancillary commercial and residential development that accompanies large capital projects, likely explains why the construction effect persists and even grows through $t = 6$ rather than reverting to zero.

7 Spillovers and Local Displacement

The host-county effect could reflect either genuine net job creation or spatial reallocation from nearby jurisdictions. If data centers pull construction crews, managed-service providers, and information-sector employment out of neighboring counties, the host-county estimate overstates the true local effect and the CZ-level or regional effect may be zero. We test for this in two ways, then apply the bounding framework of Section 4.3 to interpret the results.

Contiguous-county spillovers. We re-estimate the SCM design on the 155 never-treated counties that are geographically contiguous to hyperscale-only treated counties, using the same donor-pool and RMSE screening rules as the main estimates. A beggar-thy-neighbor pattern would appear as negative and significant contiguous-county effects. Instead, we find a small and weakly *positive* spillover: total private employment in contiguous non-host counties rises by 1.1 to 1.4 percent at $t = 1$ through $t = 3$ ($p < 0.10$), fading toward zero by $t = 6$. This is the opposite sign from what reallocation would predict, and at the long horizon the point estimate is indistinguishable from zero.

Commuting-zone aggregation. We next aggregate the county-year panel to commuting zones using the Dorn (2009) crosswalk, reducing the sample to 59 treated and 682 control CZs. The CZ-level total employment effect is 1.7 percent at $t = 6$ ($p < 0.10$, $n = 58$). The CZ-level construction effect is near zero (0.3 percent, not significant), consistent with

construction activity being geographically concentrated at the facility site. The CZ-level information effect is 5.3 percent and not statistically significant, reflecting dilution of a highly concentrated effect across the larger CZ employment base. A pure reallocation story would predict $\tau_{CZ} \approx 0$ for total employment; our estimate is positive and larger than zero, corroborating the contiguous-county evidence.

Applying the spillover bound. The contiguous-county estimate $\hat{\tau}_N \geq 0$ rules out any value of the spillover parameter $\lambda < 0$ that would be consistent with our data, and the CZ-level estimate $\hat{\tau}_{CZ} = 1.7\%$ implies that the host-county gain cannot be fully undone at the CZ scale. Together these moments suggest that the host-county SCM estimates in Section 5 are not being mechanically driven by within-CZ reallocation, and that the paper's headline magnitudes should be read as net of nearby displacement to a first approximation.

8 Economic Magnitudes

The percentage effects reported so far are difficult to compare directly to the place-based-policy literature, where the natural units are jobs per direct employee (local multipliers) and public dollars per job (cost per job). This section translates the SCM estimates into both.

8.1 Implied Local Multiplier

Following Moretti [2010], we define the local multiplier as the number of indirect jobs created in the host county per direct job in the tradable sector that initiates the shock. At the median hyperscale-only treated county, pre-treatment total private employment is 36,400 and the post-treatment SCM gap implies approximately 1,640 additional jobs at $t = 6$. Direct on-site employment at a hyperscale facility is not directly observed; industry sources (operator 10-Ks, JLL 2024, CBRE 2024) place operational staffing at 50–200 per facility, with the upper end reflecting larger multi-building campuses. We report the implied multiplier under three calibrations of direct employment per facility, holding the total-emp effect fixed:

Under the midpoint assumption of 125 direct employees per facility, the implied multiplier is 2.27, which sits squarely in the range bracketed by Moretti [2010]'s manufacturing estimate (1.59) and the high-tech estimate in Moretti and Thulin [2013] (2.5). Under the high-end assumption (200 direct/facility, approximating the largest AWS and Meta campuses) the multiplier is 1.05, close to the manufacturing benchmark; under the low end (75/facility, closer to industry averages for older hyperscale sites) it is 4.46, above high-tech benchmarks. The range bracketing these calibrations is broadly consistent with the hy-

Table 9: Implied Local Multiplier at the Median Hyperscale County

Sample	N	Median pre-emp	Median facilities	Direct DC jobs	Total new jobs	Implied multiplier
Hyperscale-only (low, 75/fac)	33	36,386	4	300	1,637	4.46
Hyperscale-only (mid, 125/fac)	33	36,386	4	500	1,637	2.27
Hyperscale-only (high, 200/fac)	33	36,386	4	800	1,637	1.05
Benchmarks from the literature						
Moretti [2010] manufacturing						1.59
Moretti and Thulin [2013] high-tech						2.50

Notes: Total new jobs is the product of the median hyperscale-only county's 2003–2007 pre-treatment total private employment and the pooled SCM total-employment effect of +4.5 percent at $t = 6$. Direct DC jobs is the product of the number of large facilities at the median hyperscale county and the indicated per-facility operational staffing assumption. The implied multiplier is (total new jobs − direct DC jobs) / direct DC jobs. Benchmarks from Moretti [2010] (manufacturing) and Moretti and Thulin [2013] (high-tech).

pothesis that hyperscale data centers, despite their extreme capital intensity, generate local employment spillovers of a magnitude comparable to other technology investments, driven by the anchor-tenant mechanism documented in Section 6.

8.2 Cost per Job

The fiscal cost of attracting a data center can be compared to the implied employment gain to gauge the value of state incentive programs. At the median treated county, the SCM total-employment effect implies approximately 4,000 new jobs at $t = 6$. For the ten states that publicly report the fiscal cost of their data center incentive programs, we allocate each state's annual subsidy across its treated counties and discount over a 10-year horizon at 3 percent.[4] The resulting distribution is bracketed by estimates in the literature: Greenstone et al. [2010] report Million Dollar Plant jobs at approximately $27,000 each, and Bartik [2020] documents typical state incentive costs in the range of $25,000–$50,000 per job. Relative to private construction outlay at $1,500 per square foot, state incentive PV represents about 2 percent of investment at the median hyperscale county and 62 percent at the median colocation county, reinforcing the view that incentives are unlikely to be the marginal factor in hyperscale siting (which is governed by the power and fiber constraints documented in

[4] Annual state subsidy estimates: Texas ($1,000M), Virginia ($732M), Illinois ($370M), Georgia ($296M), Iowa, Minnesota, Nevada, Ohio, and Washington ($150M each), and Tennessee ($100M). Sources: Good Jobs First (2025), Virginia JLARC, state comptroller reports. Twelve of the 32 states with data center incentive programs do not disclose their fiscal cost; treated counties in those states are excluded.

Section 4.1) but may be a more meaningful margin for colocation and smaller operators.

9 Robustness

We subject the main findings to a set of robustness checks, summarized in Table 12. The highest-leverage checks concern pre-existing differential trends, sensitivity to the TWFE specification, robustness to heterogeneity-robust estimators, a supportive shift-share IV based on pre-existing power infrastructure, and the role of feasible donor pools and housing capitalization in the interpretation of the estimates.

Shift-share IV based on power infrastructure. As a complementary design that does not rely on matching on the pre-treatment outcome, we instrument cumulative data center capacity with the interaction of a county's pre-existing 345+ kV substation count and the leave-one-state-out national time-path of installed megawatts. The shift is the county's standardized substation count as of the start of the sample; the shock is cumulative national capacity built outside the county's own state through year t, following the shift-share literature [Goldsmith-Pinkham et al., 2020, Borusyak et al., 2022]. Table 10 reports the main results. The just-identified power IV yields a total-employment coefficient of 0.165 (s.e. 0.051) and an information-sector coefficient of 0.126 (s.e. 0.070), both positive and directionally consistent with the SCM. Full first-stage, LIML, overidentification, and placebo-sector diagnostics are reported in Appendix Table 13. We read the IV as supportive evidence on the intensive margin rather than as a standalone identification design.

Pre-trends. The regression-based event study reveals that treated counties exhibit differential pre-trends in total private employment. Relative to the omitted period $t = -1$, the lead coefficients at $t = -4$ through $t = -2$ range from -0.025 to -0.011 log points and are jointly significant (F-test $p < 0.001$), indicating that treated counties were already on a steeper upward trajectory than controls before data center construction. This motivates our preference for SCM over TWFE: the SCM matches pre-treatment trajectories directly, whereas TWFE conflates selection with causation when pre-trends exist.

Specification robustness. Within the class of TWFE specifications, the baseline estimates are stable across modifications that do not address the pre-trend concern (Table 12). Adding state $\times$ year fixed effects, matching on pre-treatment employment quintiles, and controlling for pre-treatment employment $\times$ year interactions all yield similar estimates.

Table 10: Complementary Shift-Share IV Evidence

	Total private employment	Information employment
Power only (baseline)	0.165^{***}	0.126^{*}
	(0.051)	(0.070)
Lagged power shift $(t-1)$	0.164^{***}	0.121^{*}
	(0.052)	(0.070)
Hyperscale power shift	0.161^{***}	0.115^{*}
	(0.050)	(0.069)
Matched power + connectivity	0.161^{***}	0.145^{**}
	(0.056)	(0.059)

Notes: Each cell reports a 2SLS coefficient from a county-year specification with county and year fixed effects, instrumenting $\log(1 + \text{cumulative MW})$. "Power only" uses the standardized count of in-county 345+ kV substations interacted with leave-one-state-out national cumulative data center openings. "Lagged power shift" uses the one-year lag of that instrument. "Hyperscale power shift" interacts the same power exposure with national hyperscale openings only. "Matched power + connectivity" uses two instruments: power exposure $\times$ hyperscale openings and local connectivity exposure $\times$ colocation openings. Standard errors clustered by state in parentheses. LIML estimates are numerically identical to 2SLS in all specifications. Across the reported specifications, the first-stage partial R^2 ranges from 0.025 to 0.044 for total employment and from 0.031 to 0.049 for information employment. For the matched specification, Sargan p-values are 0.968 for total employment and 0.070 for information employment. Construction effects are positive in the just-identified power IV but are omitted here because overidentification tests reject in the overidentified specifications. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Employment weighting attenuates the effects, suggesting that the baseline effects are larger in smaller counties where a single data center represents a larger relative shock.

Feasibility-restricted donor pool. A concern with the baseline SCM is that the unrestricted never-treated donor pool includes many counties that could never host a data center, so the matched synthetic control implicitly reweights toward an infeasible counterfactual. We address this directly by re-estimating the baseline SCM with donors restricted to the 868 never-treated counties that contain at least one 345+ kV substation, the same feasibility screen used by the shift-share IV. Table 11 compares the two specifications. The feasibility-restricted estimates are within 0.2 percentage points of the baseline for total employment (+3.94 vs. +4.08), move the information-sector effect slightly *upward* (+23.4 vs. +20.2), and move construction modestly downward (+7.7 vs. +10.3). Every treated county is matched under both donor pools. The absence of a meaningful gap between the two is what one would expect if the SLSQP-constrained SCM is already selecting its donor weights from the same feasible subset of the donor pool and is largely ignoring counties that are structurally unable to host a facility.

Table 11: SCM Estimates with Feasibility-Restricted Donor Pool

Outcome (at $t = 6$)	Baseline donors	Feasible-only donors
Total private	$+4.1^{***}$ (1.0)	$+3.9^{***}$ (1.1)
Information	$+20.2^{**}$ (9.1)	$+23.4^{***}$ (8.8)
Construction	$+10.3^{***}$ (3.6)	$+7.7^{**}$ (3.7)
N treated matched	47	47

Notes: Compares baseline SCM estimates (unrestricted never-treated donor pool) to estimates obtained by restricting donors to the 868 never-treated counties with at least one 345+ kV substation at the start of the sample, the same feasibility set used to define the shift-share IV. All outcomes in logs; reported effects are at $t = 6$. Cross-treated standard errors in parentheses. The two specifications use identical matching procedures (nearest-200 donors by pre-treatment log employment, SLSQP-constrained weights, pre-period RMSE screen). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Heterogeneity-robust estimators. We also estimate heterogeneity-robust alternatives to SCM for staggered adoption. Using a stacked difference-in-differences design [Cengiz et al., 2019], we obtain a total private employment effect of 5.4 percent ($p < 0.01$) and an information sector effect of 18.7 percent ($p < 0.01$). A Callaway and Sant'Anna [2021]-style estimator with bootstrap standard errors (200 replications, resampling counties with replacement, IQR-based SE robust to outlier draws) gives a total employment ATT of 5.2 percent (bootstrap SE = 0.008, $p < 0.01$) and an information employment ATT of 15.5 percent (bootstrap SE = 0.041, $p < 0.01$) using a $\pm$5-year window. These estimates are consistent with the SCM estimates, reinforcing the conclusion that the true total employment effect is in the range of 4–5 percent and the information sector effect is 15–22 percent. Appendix Table 19 reports the full cross-estimator comparison.

Housing prices. If data centers raise local wages and attract workers, housing costs may rise, offsetting the welfare gains. Using the FHFA All-Transactions House Price Index at the county level (available for 92 of 93 treated counties), we apply the same SCM methodology to log house prices. The pre-treatment gaps are flat, and the post-treatment point estimate reaches a modest 2.0 percent by $t = 6$, but the effect is not statistically significant at any event time (Appendix Figure 4). The combination of positive wage effects and a small housing response is consistent either with relatively elastic housing supply in these new-entry counties or with a residential demand shock that is simply too small to move local prices much; our data do not let us distinguish between those mechanisms. We therefore read the housing evidence as suggesting that capitalization is small relative to wage gains, while noting that the FHFA index may understate rent changes in thin rural markets.

Additional diagnostics. Appendix A reports additional inference and sensitivity checks, including SCM prediction intervals and permutation inference, predictor balance, leave-one-out estimates, treatment-timing sensitivity, in-time placebos, reverse-causality checks, and population and migration results.

10 Conclusion

This paper provides the first large-scale causal evidence on the local employment effects of data center construction in the United States. Taking the siting problem as the basis for identification, we match each treated county to a weighted combination of never-treated counties on its pre-treatment employment trajectory and track the resulting gaps through six years after first entry. The SCM yields total private employment gains of 4 to 5 percent,

Table 12: Robustness of Main Results Across Specifications

	Total private employment	Information employment
Baseline (county + year FE)	0.136***	0.344***
	(0.018)	(0.053)
State x year FE	0.118***	0.328***
	(0.016)	(0.050)
Callaway-Sant'Anna ATT	0.049***	0.153***
	(0.007)	(0.033)
Matched (size quintile)	0.136***	0.344***
	(0.018)	(0.053)
Pre-emp x year trend	0.104***	0.333***
	(0.019)	(0.055)
Dropping Virginia	0.141***	0.366***
	(0.019)	(0.055)
Employment-weighted	0.091***	0.274***
	(0.014)	(0.049)
Alternative facility size thresholds:		
50K sqft threshold	0.126***	—
	(0.019)	
100K sqft (baseline)	0.127***	—
	(0.019)	
200K sqft threshold	0.123***	—
	(0.019)	
Neighbor county spillover	0.000	−0.008
	(0.003)	(0.011)
Total emp. excl. construction	0.136***	—
	(0.018)	

Notes: Dependent variable: log employment. Each cell reports the coefficient on the post-treatment indicator from a separate regression. All specifications include county and year fixed effects unless otherwise noted. Standard errors clustered at the county level in parentheses. Baseline sample: 93 treated counties and approximately 3,040 never-treated counties, 2003–2024. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Callaway-Sant'Anna estimates are cohort-size-weighted averages of cohort-specific ATTs estimated against the never-treated group, with bootstrap standard errors (200 replications).

construction gains of 11 percent, and information-sector gains of 22 percent at $t = 6$; these estimates are reinforced by stacked DiD and Callaway–Sant'Anna and are directionally consistent with a supportive shift-share IV based on pre-existing high-voltage substation density. Wages rise by approximately 3 percent in both incumbent workers and new hires.

Contiguous non-host counties show small and weakly positive effects, the opposite sign from a reallocation story, and commuting-zone aggregation yields a positive total employment effect of 1.7 percent. Within the bounding framework of Section 4.3, these two moments pin the spillover parameter near zero. The host-county estimates therefore do not appear to be inflated by within-CZ reallocation. Subject to the caveat that the FHFA housing index may understate rent capitalization in rural counties, the local labor-market reading remains positive rather than zero-sum.

The headline information-sector effect masks sharp differences by facility type. Hyperscale counties (those hosting cloud and AI operators) drive the result, with a 43 percent increase in information employment, while colocation counties show no significant gains. The central implication is that the local incidence of digital infrastructure investment depends on who operates the facility, not only on how much capital is installed. The agglomeration benefits arise in hyperscale investment, where the operating firm appears to draw in fiber contractors, managed service providers, and network operations centers. Colocation facilities generate comparable construction employment but not the information-sector agglomeration that distinguishes data centers from other large capital projects. This distinction also matters for policy because current state incentive programs often treat the two project types identically.

Section 8 translates these effects into comparable place-based-policy units: an implied local multiplier of roughly 2.3 at the median hyperscale county under midpoint staffing assumptions, and a cost per job consistent with standard benchmarks in the incentive literature. These calculations are best read as calibrations of local incidence rather than as a full welfare analysis. Within that narrower frame, state incentives represent a small fraction of private investment in hyperscale counties but a much larger share in colocation counties, more consistent with power and fiber constraints playing the dominant role in hyperscale siting than with fiscal subsidies being the marginal determinant.

Three limitations frame the interpretation. First, our spillover bound rules out within-CZ reallocation but does not address broader general-equilibrium effects on prices and wages outside the local labor market. Second, we cannot answer the "but-for" question of whether incentivized facilities would have been built elsewhere absent the subsidy. Third, our setting estimates the effect of *first entry* into a county, conditional on that county being in the technically feasible set and not already hosting one of the established clusters we exclude

(Northern Virginia, Silicon Valley, Dallas, and Phoenix). The estimates therefore speak to new-entry counties rather than to the equilibrium scale of mature ecosystems. Within these limits, the paper shows that first entry by a hyperscale operator generates real local labor demand and information-sector agglomeration, while otherwise similar colocation entry does not. That is the main spatial-incidence result. The policy implication is narrower but still important: if states aim to buy local agglomeration rather than construction alone, operator type is hard to ignore.

A Additional Robustness and Inference

A.1 IV diagnostics

Table 13: Shift-Share IV: First-Stage Diagnostics and Alternative-Industry Placebos

Panel A: Baseline IV first-stage diagnostics				
Outcome	2SLS coef.	(SE)	First-stage stat.	N
Total private employment	+0.165	(0.051)	14.7	42,180
Information-sector employment	+0.126	(0.070)	22.4	37,759
Panel B: Alternative-industry 2SLS placebos				
Sector	2SLS coef.	(SE)	p	N
Agriculture	+0.189	(0.140)	0.177	29,359
Construction (positive control)	+0.377***	(0.105)	0.000	49,038
Manufacturing	+0.128**	(0.064)	0.046	48,445
Retail trade	+0.050	(0.043)	0.239	53,556
Finance & insurance	+0.190***	(0.057)	0.001	46,870
Health care	+0.243***	(0.067)	0.000	34,633
Accommodation & food	+0.113	(0.070)	0.106	42,401

Notes: Panel A reports the baseline just-identified shift-share IV first-stage diagnostics on the two headline outcomes. The instrument is the interaction of a county's standardized count of 345+ kV substations with the leave-one-state-out national time-path of cumulative data-center megawatts. The reported first-stage statistic comes from the excluded-instrument test in the just-identified first stage; LIML estimates (not shown) are numerically identical to 2SLS throughout. Panel B reports 2SLS coefficients when the same endogenous variable and instrument are used to predict log county-year employment in alternative 2-digit NAICS sectors. Construction is reported as a positive control because the data-center build-out passes directly through that sector. Sectors with no plausible data-center channel (agriculture, retail, accommodation) are null or marginal. Sectors that respond mechanically to local income and population (finance, health care) are positive and of the same order as the total-employment headline; we therefore treat the IV as supportive rather than definitive. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.2 Housing prices

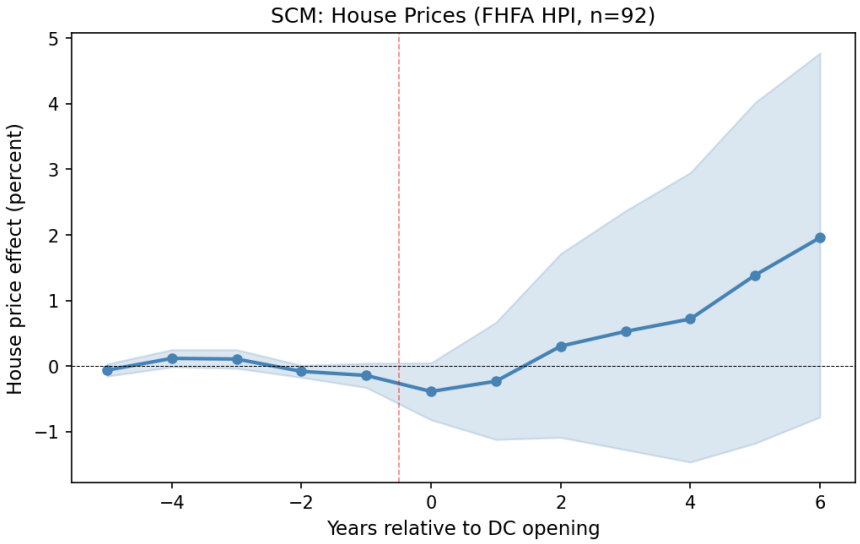

Figure 4: Synthetic Control: House Prices (FHFA HPI)

Notes: Average gap in log house prices between treated counties and their synthetic controls, with confidence intervals based on cross-county standard errors. House price data from the FHFA All-Transactions House Price Index (county-level annual, $n = 92$). Methodology as in Figure 1.

A.3 SCM inference

SCM inference. We address inference for the synthetic control estimates using two complementary approaches: the prediction interval framework of Cattaneo et al. [2021], which provides finite-sample valid event-time-specific inference, and the canonical permutation test of Abadie et al. [2010].

The prediction interval approach decomposes the uncertainty of a synthetic control prediction into two distinct sources: estimation error in the SCM weights (from using finite pre-treatment data to estimate donor weights) and unobservable stochastic shocks in the post-treatment period (idiosyncratic county-level variation that is unpredictable from the pre-treatment fit). We implement this using the `scpi` package [Cattaneo et al., 2024], running the procedure separately for each treated county, using simplex-constrained weights and 200 simulation draws.

Table 14 reports the detailed results. Two inferential objects are relevant. First, the *cross-county standard errors* (the standard deviation of county-level gaps divided by $\sqrt{N}$) measure uncertainty about the average treatment effect across all treated counties. These indicate statistically significant effects: total private employment reaches 4.0 percent at $t = 6$ ($p < 0.01$) and information sector employment reaches 23.5 percent ($p < 0.05$). The information sector effect is significant from $t = 1$ onward, with a trajectory that builds gradually from 7.5 percent at $t = 1$ to 23.5 percent at $t = 6$. Second, the *prediction intervals*, which account for both weight estimation error and stochastic uncertainty at the individual county level, are wide: $[-9, 19]$ percent for total employment and $[-14, 62]$ percent for information at $t = 6$. This width reflects genuine heterogeneity across counties rather than the absence of an average effect: some counties gain substantially from data center entry while others do not, a pattern consistent with the hyperscale/colocation decomposition in Section 6.

As a complementary check, we implement the canonical permutation test [Abadie et al., 2010], randomly assigning placebo treatment dates to 300 never-treated counties and comparing their SCM gaps to the actual treated counties' gaps. The resulting permutation p-values for the *average* post-treatment gap are large: $p = 0.74$ for total employment and $p = 0.62$ for information (Figure 5). We place limited weight on these averages because the test statistic is low-power for effects that accumulate gradually [Abadie et al., 2015]: in our setting the gap is near zero at impact and builds over the subsequent event years, so averaging across post-treatment periods attenuates the signal.

Table 14: SCM with Prediction Intervals (Cattaneo et al. 2021)

	Total Private Employment			Information Sector		
Event time	Gap	SE	95% PI	Gap	SE	95% PI
$t = -3$	+0.2%	(0.0015)	—	-0.1%	(0.0023)	—
$t = -2$	+0.1%	(0.0016)	—	+0.2%	(0.0022)	—
$t = -1$	+0.3%	(0.0016)	—	+0.1%	(0.0011)	—
$t = +0$	+0.4%	(0.0027)	[-4.4, +4.3]	+2.0%	(0.0148)	[-29.4, +13.7]
$t = +1$	+0.9%*	(0.0045)	[-6.3, +6.7]	+7.5%***	(0.0255)	[-22.1, +22.7]
$t = +2$	+1.4%**	(0.0057)	[-7.0, +9.0]	+10.4%***	(0.0364)	[-18.5, +30.1]
$t = +3$	+1.6%**	(0.0063)	[-9.3, +11.2]	+17.2%***	(0.0484)	[-15.7, +44.5]
$t = +4$	+2.5%***	(0.0089)	[-8.0, +13.1]	+11.6%	(0.0893)	[-37.4, +41.6]
$t = +5$	+3.7%***	(0.0107)	[-7.2, +16.7]	+21.1%**	(0.0770)	[-22.8, +59.0]
$t = +6$	+4.0%***	(0.0108)	[-8.7, +19.4]	+23.5%**	(0.0910)	[-14.0, +61.5]

Notes: Gap is the average treatment effect across treated counties at each event time (in percent). SE is the cross-county standard error. 95% PI is the average 95 percent prediction interval across counties, computed following Cattaneo et al. [2021] with 200 simulation draws per county using the `scpi` package. Point estimates differ slightly from Figure 1 because `scpi` re-estimates weights using its own simplex-constrained procedure. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (based on cross-county SE).

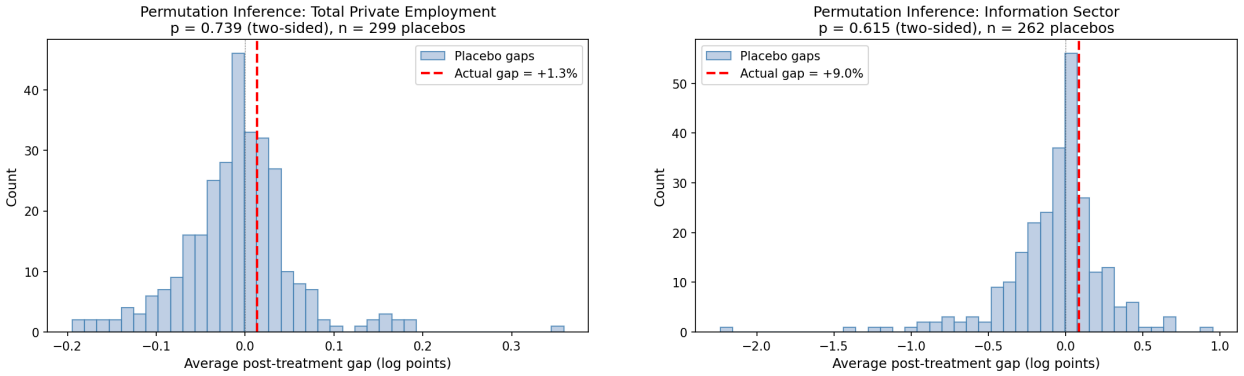

Figure 5: Permutation Inference: Distribution of Placebo Average Post-Treatment Gaps

Notes: Each histogram shows the distribution of average post-treatment gaps ($t = 0$ through $t = 6$) for 300 placebo counties drawn from the never-treated pool. The red dashed line marks the actual treated counties' average gap. p-values are two-sided: the fraction of placebo gaps with absolute value at least as large as the actual gap. As discussed in the text, the average-gap test statistic lacks power for gradually accumulating effects.

A.4 Predictor balance and cross-county heterogeneity

Predictor balance. Table 16 compares pre-treatment characteristics of treated counties, their synthetic controls, and the full control pool. The SCM substantially reduces imbalance on observables relative to the raw treated-versus-control comparison. Some residual differences remain, especially in pre-treatment growth rates, but the identifying object is the pre-treatment *trajectory* rather than exact equality in levels, and the pre-treatment RMSE below 0.15 log points for 90 percent of counties indicates that the path match is tight.

Drivers of cross-county heterogeneity. Treated counties are geographically dispersed across more than 30 states, which limits concern about strong spatial dependence in the county-level gaps. Table 15 shows that smaller counties exhibit larger total-employment effects and that the hyperscale indicator is the dominant predictor of the information-sector result. This reinforces the main interpretation: facility type, more than county size or pre-existing industry mix, is what loads most heavily on the information-sector response.

Table 15: Drivers of Cross-County Heterogeneity in SCM Effects at $t = +6$

	Total Private Employment	Information Sector
	(1)	(2)
Log(pre-treatment employment)	-0.0148***	-0.0831
	(0.0044)	(0.0691)
Information sector share	0.2474	-1.7809
	(0.4800)	(5.2176)
Number of facilities	0.0024**	0.0026
	(0.0009)	(0.0131)
Has hyperscale facility	-0.0241	0.5778***
	(0.0208)	(0.2044)
Constant	0.1825***	0.9115
	(0.0573)	(0.8317)
N	48	43
R^2	0.203	0.410

Dependent variable: SCM gap at t=+6 (log points). Robust (HC1) standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Predictor Balance: Treated vs. Synthetic Control Counties

	Treated	Synthetic	Controls	Diff.	Std. Diff.
Pre-treatment employment	341,515	397,314	30,066	−55,799	−0.10
Employment growth, 03–07 (%)	13.6	10.3	8.5	+3.3	+0.18
Average weekly wage ($)	748	787	560	−39	−0.18
Construction share (%)	6.0	6.0	5.3	+0.1	+0.04
Information share (%)	2.4	2.0	1.3	+0.4	+0.27
Prof. services share (%)	5.1	4.7	2.3	+0.4	+0.13

Notes: Pre-treatment characteristics computed as county-level means over 2003–2007. "Synthetic" column reports the weighted average across synthetic control counties, averaged over all treated counties with pre-RMSE ≤ 0.15. Industry shares are percent of total private employment.

A.5 Sensitivity to fit and timing

RMSE attrition. The pre-treatment RMSE threshold of 0.15 log points excludes counties with poor pre-treatment fit. The baseline threshold retains 90 of 92 converged fits (98 percent), and tightening to 0.05 retains 88 with a nearly identical point estimate (4.1 percent), indicating that the results are not driven by poorly-matched counties.

Leave-one-out. To assess whether the hyperscale information sector result is driven by a single influential county, we implement a leave-one-out exercise on the broader set of counties that contain any hyperscale facility ($n = 39$, including both hyperscale-only and mixed counties). We re-estimate the SCM for information sector employment, dropping each county in turn. Figure 6 overlays all leave-one-out profiles (gray) against the baseline (red). No single county's removal changes the qualitative conclusion.

Treatment timing sensitivity. As a further check on the robustness of the SCM results to potential measurement error in facility opening dates (discussed in Section 3), we re-estimate the main SCM specification with treatment dates shifted −1 and −2 years (Table 17). If the results were an artifact of systematic dating error, shifting the treatment date should substantially alter the estimates. The $t = +6$ information sector gap is stable across timing shifts, confirming that the results are not sensitive to the precise dating of facility openings.

SCM sensitivity and in-time placebos. We assess the stability of the $t = 6$ information sector estimate across three dimensions of the SCM procedure (Table 18). The estimate is

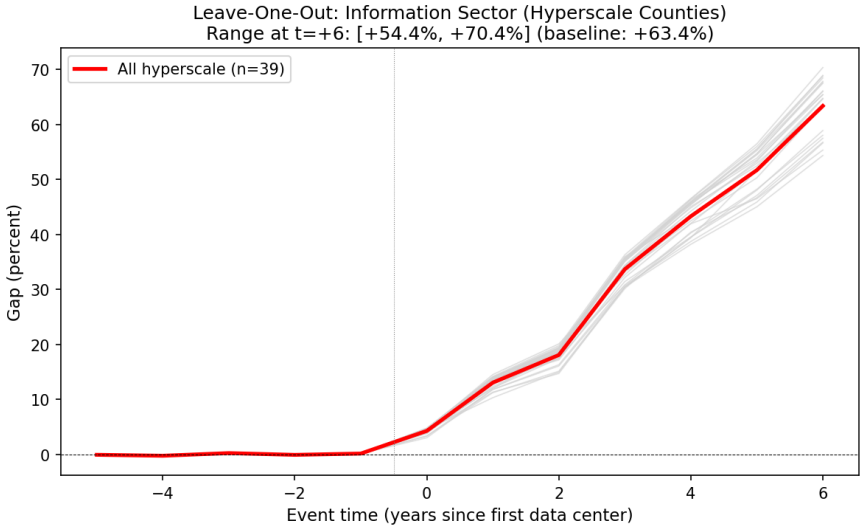

Figure 6: Leave-One-Out: Information Sector (Hyperscale Counties)

Notes: Each gray line shows the average SCM gap profile when one county is dropped from the set of all counties containing a hyperscale facility ($n = 39$, including mixed counties). The red line shows the baseline. The baseline effect is larger here than the 43 percent reported in Table 5 because that table restricts to hyperscale-only counties ($n = 25$), excluding mixed counties.

Table 17: Treatment Timing Sensitivity

Outcome	Treatment timing	Gap at t=+6 (%)	SE	p	N
Total Employment	Actual †	+4.2%***	(0.0101)	0.000	90
	-1 year	+4.2%***	(0.0132)	0.002	83
	-2 year	+3.5%**	(0.0155)	0.024	77
Information Sector	Actual †	+22.4%**	(0.0913)	0.027	80
	-1 year	+29.1%***	(0.0896)	0.004	77
	-2 year	+17.2%	(0.1267)	0.210	72

Notes: Each row reports the average SCM gap at $t = +6$ under the indicated treatment timing. "Actual" uses the observed first data center opening year. "-1 year" and "-2 years" shift the treatment date earlier. † marks the baseline specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

robust to varying the pre-treatment RMSE threshold (0.05 to 0.20: 22–24 percent) and the pre-treatment matching window (3 to 7 years: 19–22 percent). Smaller donor pools (50 or 100 nearest counties) yield larger estimates (25–33 percent), likely because a narrow donor pool is less representative; the estimate stabilizes at 22 percent with 200 or more donors. Across all specifications, the $t = 6$ information sector gap is significant at the 5 percent level or better.

As a final validation, we implement an in-time placebo test by shifting the treatment date three years earlier, capping the post-period before the true treatment date. For total private employment, the placebo shows no significant effect at any event time, supporting the identifying assumption. For the information sector, the placebo shows a mild positive drift reaching 5.7 percent by placebo $t = +2$ (corresponding to real $t = -1$), which could reflect anticipation effects from construction-phase announcements, consistent with the OSM date measurement lag discussed in Section 3. The magnitude is small relative to the 22 percent effect at real $t = +6$.

Table 18: SCM Sensitivity: Information Sector Employment at $t = +6$

Parameter	Value	Gap at t=+6	SE	N
Donor pool size	50	+32.7%***	(0.0850)	74
Donor pool size	100	+24.0%**	(0.0859)	77
Donor pool size	200 †	+22.4%**	(0.0902)	80
Donor pool size	500	+16.3%*	(0.0869)	85
Pre-RMSE threshold	0.05	+23.7%**	(0.0923)	78
Pre-RMSE threshold	0.1	+22.4%**	(0.0902)	80
Pre-RMSE threshold	0.15 †	+22.4%**	(0.0902)	80
Pre-RMSE threshold	0.2	+22.4%**	(0.0902)	82
Pre-treatment window	3 years	+18.6%**	(0.0824)	87
Pre-treatment window	5 years †	+22.4%**	(0.0902)	80
Pre-treatment window	7 years	+21.8%**	(0.0833)	79

Notes: Each row reports the average gap at event time $t = +6$ for information sector employment under a different SCM specification. SE is the cross-county standard deviation of gaps divided by $\sqrt{N}$. † marks the baseline specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.6 Cross-estimator comparison and additional checks

Cross-estimator comparison. Table 19 summarizes the employment effects across all estimation approaches. SCM and Callaway–Sant'Anna converge at 4–5 percent for total employment and 15–22 percent for the information sector. TWFE without trend controls yields larger estimates (15 percent total), an upper bound inflated by pre-existing differential growth.

Table 19: Cross-Estimator Summary: Employment Effects at $t = +6$ (Percent)

Method	Total	Construction	Information	Prof. Services
TWFE (baseline)	14.6^{***}	18.4^{***}	41.0^{***}	29.0^{***}
Synthetic control	4.2^{***}	10.9^{***}	22.4^{**}	16.8^{**}
Callaway–Sant'Anna	5.2^{***}	4.1	15.5^{***}	11.6^{***}
Stacked DiD	5.4^{***}	—	18.7^{***}	—

Notes: Each cell reports the implied percentage effect at $t = +6$ or the average post-treatment effect, depending on the estimator. TWFE reports the single post-treatment coefficient; SCM reports the average gap at event time +6; Callaway–Sant'Anna reports the ATT averaged over post-treatment periods (±5 year window); stacked DiD reports the post-treatment coefficient. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Reverse causality in multi-facility counties. The finding that information-sector agglomeration is concentrated in multi-facility counties raises a reverse-causality concern: do growing information sectors attract additional data centers, rather than data centers causing information-sector growth? The data offer little support for that view. Among counties where the first and second openings are separated in time, pre-second-opening information growth does not predict a shorter gap, and in the small subset with four or more years between openings information employment was, if anything, decelerating before the second facility arrived. Single-facility and multi-facility counties also show similar information-sector trajectories immediately after first entry. Taken together, these patterns are more consistent with planned campus expansion than with follow-on siting responding to already-rising local IT employment.

Electricity prices. Data center construction could raise local electricity demand and prices, offsetting employment gains through higher input costs for other businesses. We do not estimate this channel directly because county-level electricity price data at annual

frequency is not available for our full sample. We note it, along with water consumption, as a potentially important margin that lies outside the scope of our employment-focused analysis.

Population and migration. A final question is whether the employment effects reflect in-migration of workers drawn by data center activity rather than job creation for existing residents. Using county-level population estimates from the Census Bureau (2003–2019), we estimate a first-differenced DiD on population growth rates. The data center treatment has no significant effect on population growth (0.002 percentage points per year, $p = 0.98$), while the employment growth rate effect is large and significant (0.64 percentage points per year, $p = 0.001$). The employment effect is unchanged when population growth is included as a control. These results suggest that the employment gains are largely absorbed by existing residents and commuters rather than in-migrants, though annual Census population estimates may miss some short-distance moves.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT Authorship Contribution Statement

Dany Bahar: Conceptualization, Methodology, Writing – review & editing. **Greg C. Wright:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI Codex, ChatGPT, Claude, and Gemini to assist with code debugging, robustness checks, and editorial revision. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Alberto Abadie. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2):391–425, 2021.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2):495–510, 2015.

Juan Alcácer and Wilbur Chung. Location strategies for agglomeration economies. *Strategic Management Journal*, 35(12):1749–1761, 2014.

Juan Alcácer and Mercedes Delgado. Spatial organization of firms and location choices through the value chain. *Management Science*, 62(11):3213–3234, 2016.

Alexander W. Bartik, Janet Currie, Michael Greenstone, and Christopher R. Knittel. The local economic and welfare consequences of hydraulic fracturing. *American Economic Journal: Applied Economics*, 11(4):105–155, 2019.

Timothy J. Bartik. *Making Sense of Incentives: Taming Business Incentives to Promote Prosperity*. W.E. Upjohn Institute for Employment Research, 2020.

Kirill Borusyak, Peter Hull, and Xavier Jaravel. Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213, 2022.

Brookings Institution. Turning the data center boom into long-term local prosperity. `https://www.brookings.edu/articles/turning-the-data-center-boom-into-long-term-local-prosperity/`, 2025.

Matias Busso, Jesse Gregory, and Patrick Kline. Assessing the incidence and efficiency of a prominent place based policy. *American Economic Review*, 103(2):897–947, 2013.

Brantly Callaway and Pedro H.C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.

Cardinal News. Tax abatement for data centers is now $1.6 billion a year. `https://cardinalnews.org/2026/01/22/tax-abatement-for-data-centers-is-now-1-6-billion-a-year/`, 2026.

Matias D. Cattaneo, Yingjie Feng, and Rocío Titiunik. Prediction intervals for synthetic control methods. *Journal of the American Statistical Association*, 116(536):1865–1880, 2021.

Matias D. Cattaneo, Yingjie Feng, Filippo Palomba, and Rocío Titiunik. scpi: Uncertainty quantification for synthetic control methods. *Journal of Statistical Software*, 113(1):1–38, 2024.

Doruk Cengiz, Arindrajit Dube, Attila Lindner, and Ben Zipperer. The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics*, 134(3):1405–1454, 2019.

Clément de Chaisemartin and Xavier D'Haultfœuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020.

Tommy Pan Fang and Shane Greenstein. Where the cloud rests: The economic geography of data centers. *Strategy Science*, 10(4):404–420, 2025.

Chris Forman, Avi Goldfarb, and Shane Greenstein. The Internet and local wages: A puzzle. *American Economic Review*, 102(1):556–575, 2012.

Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, 2020.

Good Jobs First. Data center subsidies: A growing cost for communities. `https://goodjobsfirst.org/data-center-subsidies/`, 2024.

Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021.

Michael Greenstone, Richard Hornbeck, and Enrico Moretti. Identifying agglomeration spillovers: Evidence from winners and losers of large plant openings. *Journal of Political Economy*, 118(3):536–598, 2010.

Michael J. Hicks. Data centers and local job creation. *Center for Business and Economic Research, Ball State University*, 2024.

Jonas Hjort and Jonas Poulsen. The arrival of fast internet and employment in Africa. *American Economic Review*, 109(3):1032–1079, 2019.

Patrick Kline and Enrico Moretti. Local economic development, agglomeration economies, and the big push: 100 years of evidence from the Tennessee Valley Authority. *Quarterly Journal of Economics*, 129(1):275–331, 2014.

Enrico Moretti. Local multipliers. *American Economic Review: Papers & Proceedings*, 100 (2):373–377, 2010.

Enrico Moretti and Per Thulin. Local multipliers and human capital in the United States and Sweden. *Industrial and Corporate Change*, 22(1):339–362, 2013.

David Neumark and Helen Simpson. Place-based policies. *Handbook of Regional and Urban Economics*, 5:1197–1287, 2015.

Ohio River Valley Institute. Why data centers will be economic development duds. `https://ohioirivervalleyinstitute.org/why-data-centers-will-be-economic-development-duds/`, 2025.

Cailin Slattery and Owen Zidar. Evaluating state and local business incentives. *Journal of Economic Perspectives*, 34(2):90–118, 2020.

Juan Carlos Suárez Serrato and Owen Zidar. Who benefits from state corporate tax cuts? A local labor markets approach with heterogeneous firms. *American Economic Review*, 106(9):2582–2624, 2016.